

Validation of AI-generated results in theoretical physics

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AI now generates physics results faster than we can check them. A central part of the answer: machine-checkable certificates, not human reading alone. A survey of AI for mathematics and physics, and of formalization as the validation technology.

Contents

1. The validation problem
2. Recent progress: AI for mathematics and physics
3. What is validation?
4. The validation toolbox
5. Is physics ready?
6. It is real
7. Challenges and outlook

The validation problem

A scenario

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Understanding the program well enough to rely on it will take us **far longer** than it took the AI to write it. Our ability to **validate** has become the bottleneck.

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- Humans have protocols for this – but they do not transfer to AI, and the goal is still to convince *humans* of correctness.

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An **algorithm-checkable certificate** – a deterministic, near-100%-reliable check, cheaper than the derivation (*not* an LLM checking an LLM) – must become a *central component* of trust: the part that scales when we can no longer read the output by hand.

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What is new is not the idea of checking – it is that AI makes *generation* so cheap that **checking, not generating, becomes the bottleneck.**

Recent progress: AI for mathematics and physics

The state of AI, 2025–2026

- LLMs were optimized for next-word prediction (through GPT-4); performance jumped with RL on new reward signals – reasoning, coding, tool use.
- Roughly 100 Elo points/year on public arenas; early-2026 models (Claude Opus 4.5+, Gemini 3 Pro, GPT-5) far stronger than 2024.
- Excellent at retrieval and adapting known arguments; weaker at original logical reasoning – much current work uses search, verification and self-check to compensate.
- Strongest in high-resource domains (Python); weaker in low-resource ones (most academic specialties).

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The frontier crossed a threshold for research-level math in late 2025.

AI for mathematics – impressive, but unverified

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Whether AI can *do* mathematics is a deep question in its own right – but the question for *this* talk is “**how do we know the answer is right?**”

Background – interactive theorem provers and Lean

The “verified” results that follow rely on one tool, new to most physicists. An **interactive theorem prover (ITP)** is a language for stating definitions and theorems and *proving* them, where a **fixed algorithm** checks every step – so a checked proof is essentially 100% reliable, and **interpretable**. You work at a high level (via “tactics”), much as on paper. A proof looks like:

```
1 theorem infinite_primes (n : ℕ) : ∃ p, n ≤ p ∧ Prime p :=
2   let p := minFac (n ! + 1)
3   have f1 : n ! + 1 ≠ 1 := ne_of_gt <| succ_lt_succ <| factorial_pos _
4   have pp : Prime p := minFac_prime f1
5   have np : n ≤ p :=
6     le_of_not_ge fun h =>
7       have h1 : p | n ! := dvd_factorial (minFac_pos _) h
8       have h2 : p | 1 := (Nat.dvd_add_iff_right h1).2 (minFac_dvd _)
9       pp.not_dvd_one h2
10  ⟨p, np, pp⟩
```

Lean and mathlib – what has been formalized

Lean 4 is the leading system for mathematics; its library **mathlib** ($\sim 2\text{M}$ lines) spans much of the undergraduate curriculum and well beyond. Landmarks:

- Four-Colour and Feit–Thompson theorems (Gonthier *et al*); the Kepler conjecture (Hales, Flyspeck).
- Scholze–Clausen liquid tensor experiment (Commelin *et al*); polynomial Freiman–Ruzsa (Gowers, Green, Manners, Tao); Carleson’s theorem (2025).
- In progress: Fermat–Wiles (Buzzard). Physics formalization is just beginning (more shortly).

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A growing community uses Lean – and, increasingly, AI agents write it, removing the cost that for two decades made this a specialist’s tool. The point to carry forward: a Lean-checked proof is a **machine-checkable certificate**.

AI for mathematics – now verified

AlphaProof Nexus (DeepMind, arXiv:2605.22763, May 2026): LLM agents + Lean.

- Solved **9 of 353** open Erdős problems (one open 56 years), **44 of 492** OEIS conjectures, a 15-year-old algebraic-geometry question.
- $\sim$ a few hundred dollars per problem.
- Every proof is mechanically checked by Lean – *no expert review of the logic needed*.

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(Humans still vet that the formal *statement* captures the conjecture – hold that thought.)

Certificate by construction

AlphaEvolve (DeepMind, 2025) and **FunSearch**: evolutionary search where a candidate is kept only if it *passes an exact evaluator*.

- 4×4 complex matrix multiply in 48 multiplications – first improvement on Strassen in 56 years.
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Here the result is **trustworthy by construction**: the certificate *is* the check the search optimizes against. The purest form of “trust the output, not the generator.”

Scale – and a skeptic

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Tao’s caution is worth quoting: automation first harvests the *lowest-hanging fruit*; formalization is what keeps the harvest *honest*.

AI for physics (1) – generating results

AI is now producing *physics*, not just mathematics. Two Harvard examples:

- **Scattering amplitudes** (Schwartz, Cheung, Dersy; IAI/Fl). Transformers predict coefficients in the amplitude bootstrap for planar $\mathcal{N} = 4$ super Yang–Mills (arXiv:2405.06107); symbolic regression “learns the simplicity” of amplitudes, distilling huge expressions to compact closed forms (arXiv:2408.04720).
- **Discovery by program search** (Brenner; Harvard & Google). **FunSearch** – an LLM proposing programs, scored by an evaluator – found record cap sets and new constructions; related work learns PDE discretizations and effective models.

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The validation question, attached to each: a transformer’s amplitude coefficient $\rightarrow$ must satisfy bootstrap / known-limit constraints; a distilled formula $\rightarrow$ a closed form you can *check*; FunSearch $\rightarrow$ kept only if the *exact evaluator* passes.

AI for physics (2) – and where the certificate is missing

Many-body and geometry, where the check is numerical:

- **Neural quantum states:** variational ansätze for lattice systems – the *variational energy upper bound* is a built-in certificate.
- **ML for lattice QCD:** gauge-equivariant nets, flow-based sampling; validated by standard Monte-Carlo cross-checks.
- **Symbolic distillation of Calabi–Yau metrics** (arXiv:2602.07834): a 5-term formula, $R^2 = 0.9994$, validated against *physical observables* (volumes, Yukawa couplings).

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The trustworthy results are exactly the ones **carrying a cheap check**. The danger case – a fluent LLM “derivation” with *no* certificate – is the one physics culture is least equipped to catch.

AI for physics (3) – the validation showcase

The generalized quantum Stein's lemma, formalized in Lean (Meiburg *et al*, arXiv:2510.08672).

- A central theorem of quantum information – the operational meaning of relative entropy, and the reversibility of quantum resources.
- Its original proof (2010) was found to have a **gap** (Quantum 2023); re-proved by Hayashi–Yamasaki (2024).
- Now **machine-verified** – the most technically demanding physics theorem formalized to date – and the formalization *itself rectified further imprecisions* in the proof.

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gap → reproof → machine-checked → caught more

This arc is the whole thesis: formalization does not just certify – it **finds the errors**.

AI for physics (3, cont.) – the formalization landscape

- **PhysLean** (Tooby-Smith *et al*): Wick's theorem, the Higgs potential, Maxwell's equations, Lorentz-group index notation.
- **Quantum information** in Lean – the library built for the Stein result.
- **Our QFT work** – OSforGFF (4D free QFT, all OS axioms) and a lattice Yang–Mills mass gap – verified mathematical physics; details shortly.

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Small but real, and growing fast. Almost all of it formalizes *rigorous* physics – the hard, unmapped territory is everything else, which is most of physics.

Survey takeaway

Across math and physics, the AI results we *trust* are exactly the ones that came with a **check**.

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Physics has the **least-developed certificate machinery** of any quantitative field. That is the problem – and the opportunity.

What is validation?

What “validation” means

Two things at once:

- the result is **correct**, and
- we can **convince** the users/stakeholders that it is.

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The enabling fact is an **asymmetry**: checking can be far cheaper than finding.

- **SAT**: a satisfying assignment is its own certificate – trivial to check, hard to find.
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Validation technology = systematically attaching **cheap, algorithm-checkable certificates** to expensive results.

The validation toolbox

Verified symbolic computation

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`a = <defn 1> ; b = <defn 2>` vs. `b = <defn 1> ; a = <defn 2>`

(with later references to `a`, `b` unchanged).

The CAS **does not know or care** which is right. An ITP does – the types must match.

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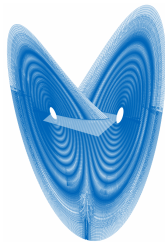
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Opportunity: have the CAS emit a *certificate* of its steps that Lean can check – keeping CAS speed, gaining ITP soundness.

Verified numerical computation

Approximate, yet rigorous: round *outward* and propagate.

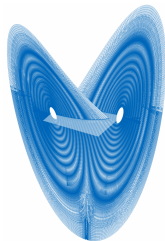
- Interval arithmetic: replace x by rigorous bounds $[a, b]$, propagate through every op.
- **Lorenz attractor** is chaotic (Tucker 2002; verified in Isabelle/HOL by Immler 2018).
- **Verified Calabi–Yau metric**: neural net finds an approximate solution; rigorous spectral-gap and error bounds + a fixed-point theorem $\Rightarrow$ an exact solution exists.



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Honest caveat: verified numerics inside ITP is still thin. Near-term: couple validated computation to an ITP so there is at least a **specification**.

Is physics ready?

Act 1 – what can be formalized at all?

A natural criterion for *scope*, not for trust: a result can be formalized iff it has a **mathematically well-defined formulation** – *not* a rigorous justification.

- Setting a Lagrangian term to zero: not rigorous, but well-defined – formalizable.
- The path integral's *steps* are well-defined, even where the object is not.
- “Approximate 1200 by 1000 but not by 800”: hard to even formulate.

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This says what is *in range* – not what is *correct*. Well-definedness is **necessary but nowhere near sufficient for trust**: one can formalize the steps of a path-integral calculation and certify *nothing* about the physics. Closing that gap is the rest of this section.

Act 2 – rigorous, or merely well-defined?

Two regimes need different paradigms:

- **Verified mathematical physics** – rigorous results; the mathematician's paradigm applies directly (stability of matter, CPT, our OS construction).
- **Verified physical computation** – collider amplitudes, bootstrap bounds, dynamical-systems bounds; needs verified CAS + numerics + a specification.

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⇒ the Standard-Model amplitudes experimentalists actually use are a realistic target. But a rigorous *derivation* is still not a trusted *result* – that turns on the specification.

Act 3 – formalizable $\neq$ validated (the gap)

A formal proof certifies an *implication*:

$$A, B, C \implies X, Y, Z.$$

It does **not** certify that A, B, C and X, Y, Z mean what we intend.

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The specification / meaning gap

The real work – and the real research frontier – is writing formal specifications that **human experts vet** as capturing the physics. This is what everyone else skips.

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Yes – and that *is* the point. Formalization does not eliminate trust; it **relocates and shrinks** it.

- **Before:** trust a 1000-page AI derivation you cannot read.
- **After:** trust a *short, explicit specification* + a tiny proof kernel. Everything in between is machine-checked.

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The specification is small, stable, reusable, and **independently auditable** – including against physics' own battery: symmetries, limiting cases, dualities, numerics.

So what does formalization actually buy you?

The skeptic's objection: haven't we just moved the trust from the proof to the spec?

Yes – and that *is* the point. Formalization does not eliminate trust; it **relocates and shrinks** it.

- **Before:** trust a 1000-page AI derivation you cannot read.
- **After:** trust a *short, explicit specification* + a tiny proof kernel. Everything in between is machine-checked.

The specification is small, stable, reusable, and **independently auditable** – including against physics' own battery: symmetries, limiting cases, dualities, numerics.

Trust = a formal certificate of the derivation + a specification that survives the physicist's consistency checks. Not certainty – a drastic, auditable reduction of what you must take on faith.

Worked example – the collider amplitude, made precise

Back to the opening: the amplitude for $g g \rightarrow W + 6 g$ at the LHC.

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What would a **specification** – the textbook definition we already trust – say?

- $\mathcal{M}$ is the order- α_s^k term of the Standard-Model perturbative expansion for this external state – which *can* be made rigorous order-by-order (cf. Costello; we need not adopt his framework).
- It must satisfy the physics that pins it down:
 - **Gauge invariance**: replace a gluon polarization $\epsilon_i \rightarrow p_i$ and the amplitude must vanish (Ward identity).
 - **Bose symmetry** under exchange of identical gluons.
 - Correct **soft** / **collinear** (infrared) singularities and factorization.

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These consistency conditions are not decorations – they are most of the specification.

Worked example – validating the 1000-page program

The AI writes the program; the machine checks it against the spec.

- **Verified symbolic computation:** the colour / Lorentz / Dirac algebra and the Feynman-rule bookkeeping emit Lean-checkable certificates.
- **Verified numerics:** the loop / phase-space integration with rigorous (interval) error bounds, or post-verified.
- **Physics consistency checks**, run automatically: Ward identity = 0; agreement with known lower-multiplicity and soft limits; gauge-parameter independence.

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The bottleneck from the opening – dissolved.

The deep frontier – foundations for physical reasoning

Physicists rarely write rigorous proofs, yet the community reaches **stable agreement** on what is established. Perhaps physical reasoning, at its best, is already rigorous – just not grounded in ZFC.

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A program: make those foundations explicit and formal.

- Formalize axiomatic QM/QFT; verify that a claimed observation conflicts with them (e.g. a CPT violation – which axiom must give?).
- Admit well-supported but unproven statements as **axioms** – asymptoticity of perturbation series, S -duality, AdS/CFT – and let the verifier **detect contradiction**.

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Explicit axioms do not make the conjectures *true* – they make the **dependency structure** checkable: which results rest on which conjectures, and where a surprise would propagate. This is where validation of AI-generated physics becomes genuinely new.

Example – codifying the lore behind an exact result

Formalizing *rigorous* physics (our free-QFT and lattice Yang–Mills results) is the mature end. The deeper frontier is the *non-rigorous* reasoning physicists trust anyway.

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Seiberg–Witten (1994): the *exact* low-energy solution of $\mathcal{N} = 2$ $SU(2)$ super Yang–Mills. The derivation is not rigorous; its physical “axioms”:

- holomorphy of the prepotential (from $\mathcal{N} = 2$ supersymmetry);
- the Coulomb branch has *exactly* the expected singularities – massless monopoles / dyons;
- electric–magnetic duality: the monodromies must compose consistently;
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Why it is trusted: **independent agreement** across perturbation theory, semiclassics / instantons (Nekrasov’s instanton sum reproduces the prepotential), and the exact answer.

Codify those axioms $\Rightarrow$ the reasoning becomes **auditable**; a verifier checks the chain and flags contradiction. The pert. / instanton / exact triangle, made formal.

It is real

Evidence (1) – a verified construction of QFT

OSforGFF (with Hoback, Mei, Nissim): formalize the 4D massive free Euclidean QFT and verify the Osterwalder–Schrader axioms.

- Construct an infinite-dimensional Gaussian measure with covariance $C(x, y) = \int d^4k e^{ik(x-y)} / (k^2 + m^2)$.
- **All five OS axioms + ergodicity. 0 sorries, 0 axioms.** ~32,000 lines.
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A physics result, certified to the same standard as formalized mathematics.

Evidence (2) – a Yang–Mills mass gap, and the pace of change

lgt (with Rajasekaran): $U(n)$ Wilson lattice Yang–Mills, exponential correlation decay at strong coupling via Dobrushin uniqueness.

- Mass-gap theorems fully proved, **0 project axioms**.
- Eight sorry-free layers: lattice geometry $\rightarrow$ Gibbs/DLR $\rightarrow$ Dobrushin decay $\rightarrow$ mass gap.

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And the trajectory: **Bochner's theorem** was proved by Claude Code in ~ 2 days, *nearly autonomously*, in February 2026. A year earlier this was unthinkable.

Challenges and outlook

The hard parts

- **The specification gap** again: who writes, and who vets, the formal statement of “the physical scattering amplitude”?
- **Who validates the validator?** An *LLM* judge can be reward-hacked (Goodhart); a deterministic proof *kernel* cannot – it is small, fixed, and auditable. The residual risk collapses onto the **specification**.
- **The insidious failure:** models led astray by nearly-but-not-quite-right analogies – complex-matrix results applied over a Grassmann algebra; conditionally convergent treated as absolutely convergent.

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- **The insidious failure:** models led astray by nearly-but-not-quite-right analogies – complex-matrix results applied over a Grassmann algebra; conditionally convergent treated as absolutely convergent.

Countermeasures that work today: cross-model validation; unit tests for new definitions; back-chaining through vetted “textbook axioms.”

Outlook

- The bottleneck in knowledge formation is shifting from discovery and writing to **selecting, reading and validating**.
- For physics: build the certificate machinery – verified CAS, verified numerics, formalized specifications, libraries of formalized physics.
- For the ML community: **autoformalization**, **certificate-emitting CAS**, and **verified numerics in ITP** are rich, underworked, open problems.

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- For the ML community: **autoformalization**, **certificate-emitting CAS**, and **verified numerics in ITP** are rich, underworked, open problems.

The question is no longer “can it be done,” but “what should we validate next?”

Conclusions

- AI generates physics faster than we can check it; **machine-checkable certificates** must become a central component of trust – alongside expert-vetted specifications and physics' own consistency checks.
- Formalization – formal proof, verified symbolic, verified numerical computation – is the validation technology, and it is **working now** at research level.
- Physics is more ready than it looks: you need not solve the foundations to validate a calculation.
- The frontier is **meaning**: formalizable is not validated. Specifications, vetted by experts, are the real work – and where “foundations for physical reasoning” could make this genuinely new.